MPU6050 Motion_Detection的使用

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项目要求，需要感知震动，感知震动后通过外部中断唤醒MCU计数以实现低功耗。
根据DATASHEET中所提，Motion_Detection需要配置*MOT_THR*、*MOT_DUR*、*MOT_DETECT_CTRL*、*MOT_DETECT_STATUS（非必须，我没有配置）*，与*Zero_Motion*一样，通过读取 *MOT_DETECT_STATUS* 来判断是XYZ哪个方向的震动。
而外部中断则需要通过配置 *INT_ENABLE* 来实现。
强调一下：

> *MOT_DETECT_STATUS*
> *MOT_THR*
> *MOT_DUR*
> *MOT_DETECT_CTRL*（非必须）

下面解释下相关寄存器的含义：

MOT_DETECT_STATUS：

61	97	MOT_DETECT_STATUS	R	MOT_XNEG	MOT_XPOS	MOT_YNEG	MOT_YPOS	MOT_ZNEG	MOT_ZPOS	-	MOT_ZRMOT

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从DATASHEET可以看出，NEG置为一，震动为沿轴负方向，POS置为一，震动为沿轴正方向。

MOT_XNEG	This bit automatically sets to 1 when motion in the negative X axis has generated a Motion detection interrupt.
MOT_XPOS	This bit automatically sets to 1 when motion in the positive X axis has generated a Motion detection interrupt.
MOT_YNEG	This bit automatically sets to 1 when motion in the negative Y axis has generated a Motion detection interrupt.
MOT_YPOS	This bit automatically sets to 1 when motion in the positive Y axis has generated a Motion detection interrupt.
MOT_ZNEG	This bit automatically sets to 1 when motion in the negative Z axis has generated a Motion detection interrupt.
MOT_ZPOS	This bit automatically sets to 1 when motion in the positive Z axis has generated a Motion detection interrupt.
MOT_ZRMOT	This bit automatically sets to 1 when Zero Motion detection interrupt is generated.

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MOT_ZRMOT为Zero_Motion位。

注：读取后中断取消，MOT_DETECT_STATUS归零

MOT_THR 与MOT_DUR：

1D	29	FF_THR	R/W	FF_THR[7:0]
1E	30	FF_DUR	R/W	FF_DUR[7:0]
1F	31	MOT_THR	R/W	MOT_THR[7:0]
20	32	MOT_DUR	R/W	MOT_DUR[7:0]
21	33	ZRMOT_THR	R/W	ZRMOT_THR[7:0]
22	34	ZRMOT_DUR	R/W	ZRMOT_DUR[7:0]

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MOT_THR表示其对震动感知的灵敏度，Product Specify上写为 *32* mg/LSB。数值越小，对震动的感知越灵敏。
MOT_DUR为震动持续时间，只有 震动强度>MOT_THR 持续时间大于等于MOT_DUR，中断才会触发。MOT_DUR的单位为 1ms/LSB。
FF为FreeFalling相应的寄存器。
ZRMOT为Zero_Motion相应寄存器。

MOT_DETECT_CTRL：

69	105	MOT_DETECT_CTRL	R/W	-	-	ACCEL_ON_DELAY[1:0]	FF_COUNT[1:0]	MOT_COUNT[1:0]

这个没有研究，相应的描述放在下面。

Description:

This register is used to add delay to the accelerometer power on time. It is also used to configure the Free Fall and Motion detection decrement rate.

The accelerometer data path provides samples to the sensor registers, Motion detection, Zero Motion detection, and Free Fall detection modules. The signal path contains filters which must be flushed on wake-up with new samples before the detection modules begin operations. The default wake-up delay, of 4ms can be lengthened by up to 3ms. This additional delay is specified in *ACCEL_ON_DELAY* in units of 1 LSB = 1 ms. The user may select any value above zero unless instructed otherwise by InvenSense. Please refer to Section 8 of the MPU-6000/MPU-6050 Product Specification document for further information regarding the detection modules.

Detection is registered by the Free Fall detection module or the Motion detection module after accelerometer measurements meet their respective threshold conditions over a specified number of samples. When the threshold conditions are met, the corresponding detection counter increments by 1. The user may control the rate at which the detection counter decrements when the threshold condition is not met by configuring *FF_COUNT* and *MOT_COUNT*. The decrement rate can be set according to the following table:

FF_COUNT or MOT_COUNT	Counter Decrement
0	Reset
1	1
2	2
3	4

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INT_ENABLE：

38	56	INT_ENABLE	R/W	FF_EN	MOT_EN	ZMOT_EN	FIFO_OFLOW_EN	I2C_MST_INT_EN	-	-	DATA_RDY_EN

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中断开启标志。允许什么事件触发中断。
讲一下不常用的：

> FF_EN：为Free_Falling中断使能。
> MOT_EN：为Motion_Detect中断使能。
> ZMOT_EN：为Zero_Motion_Detect中断使能。

好，接下来是喜闻乐见的代码说明时间

```
void MPU6050_Config()
{
    I2C_WriteRegBytes(I2C2,SlaveAddress,PWR_MGMT_1,0x08);
    I2C_WriteRegBytes(I2C2,SlaveAddress,PWR_MGMT_2,0xF7);//我关闭了陀螺仪、加速度计的X、Y轴， 高2位应该只对MPU9150有用，我设置成11
    I2C_WriteRegBytes(I2C2,SlaveAddress,SMPLRT_DIV,0x07);
    I2C_WriteRegBytes(I2C2,SlaveAddress,CONFIG,0x04);
    I2C_WriteRegBytes(I2C2,SlaveAddress,ACCEL_CONFIG,0x1C);//2g量程，这一步一定不能省略，否则灵敏度降低
    I2C_WriteRegBytes(I2C2,SlaveAddress,0x1F,0x01);//1mg triger，如果设置为0，则静止也会随机触发中断
    I2C_WriteRegBytes(I2C2,SlaveAddress,0x20,0x01);//1ms lasting，如果设置为0，则静止也会随机触发中断
    I2C_WriteRegBytes(I2C2,SlaveAddress,0x38,0x40);//MOT_EN set 1，开启感应中断
}

```

这便是MPU6050 Motion_Detect的初始化。
Zero_Motion_Detect的初始化应该跟这个套路相同。
下面是触发中断后的处理：

```
void MPU6050_Read()
{
    I2C_ReadBytes(I2C2,SlaveAddress,MOT_DETECT_STATUS,1,(uint8_t*)&MOT_Status,MSB);//按照MSB方式读取一个字节。MOT_Status即为得到的相应的数据。
}
```